Quantitative Methods I

Week 1 - Linear & Quadratic Functions

Jan Christopher Kops

QM I, SBE, Maastricht University

Why Maths?

- Mathematics provides a fundamental framework
  - for understanding the world and
  - solving problems
- Economics and Business heavily rely on maths
- And...
  - math skills and income are positively correlated

Motivation

To see why math is important and to motivate today's topic,

- consider a firm with two production technologies
  - $f(x) = 2x$ a linear function
  - $g(x) = x^2$ a quadratic function
- To boost production (i.e., to increase $f + g$),
  - the firm decides to double its input x
  - but which production technology should it use?

    a) f
    b) g
    c) f and g

Function(s) on $\mathbb{R}$

Definition

Let D, R be sets of real numbers. A function f assigns to each element $x \in D$ one element $y = f(x) \in R$.

- x is the independent or exogenous variable
- y is the dependent or endogenous variable

Function(s) on $\mathbb{R}$ - cont'd

Definition

Let D, R be sets of real numbers. A function f assigns to each element $x \in D$ one element $f(x) \in R$.

- Domain D: set of admissible real values for x
- Range R: set of real values $f(x)$ for $x \in D$
  (the set of all values $y \in \mathbb{R}$ for which $f(x) = y$ has a solution x)

Domain of a function

Consider

$$f(x) = 2\sqrt{(x-1)(5-x)}$$

- What is the domain of this function?
  - $D = [1, 5]$ is the set of admissible real values here for x
  - Why is that?
- What is its range?
  - $R = [0, 4]$ is the set of real values $f(x)$ for $x \in D$
  - Why is that?

Examples

Examples of functions in economics and business

- demand & supply functions
- cost functions
- production functions
- profit functions
- utility functions
- expected utility functions

Mathematical forms of these functions

- linear, quadratic, polynomial, exponential, logarithmic

Example - (Linear) Supply & Demand

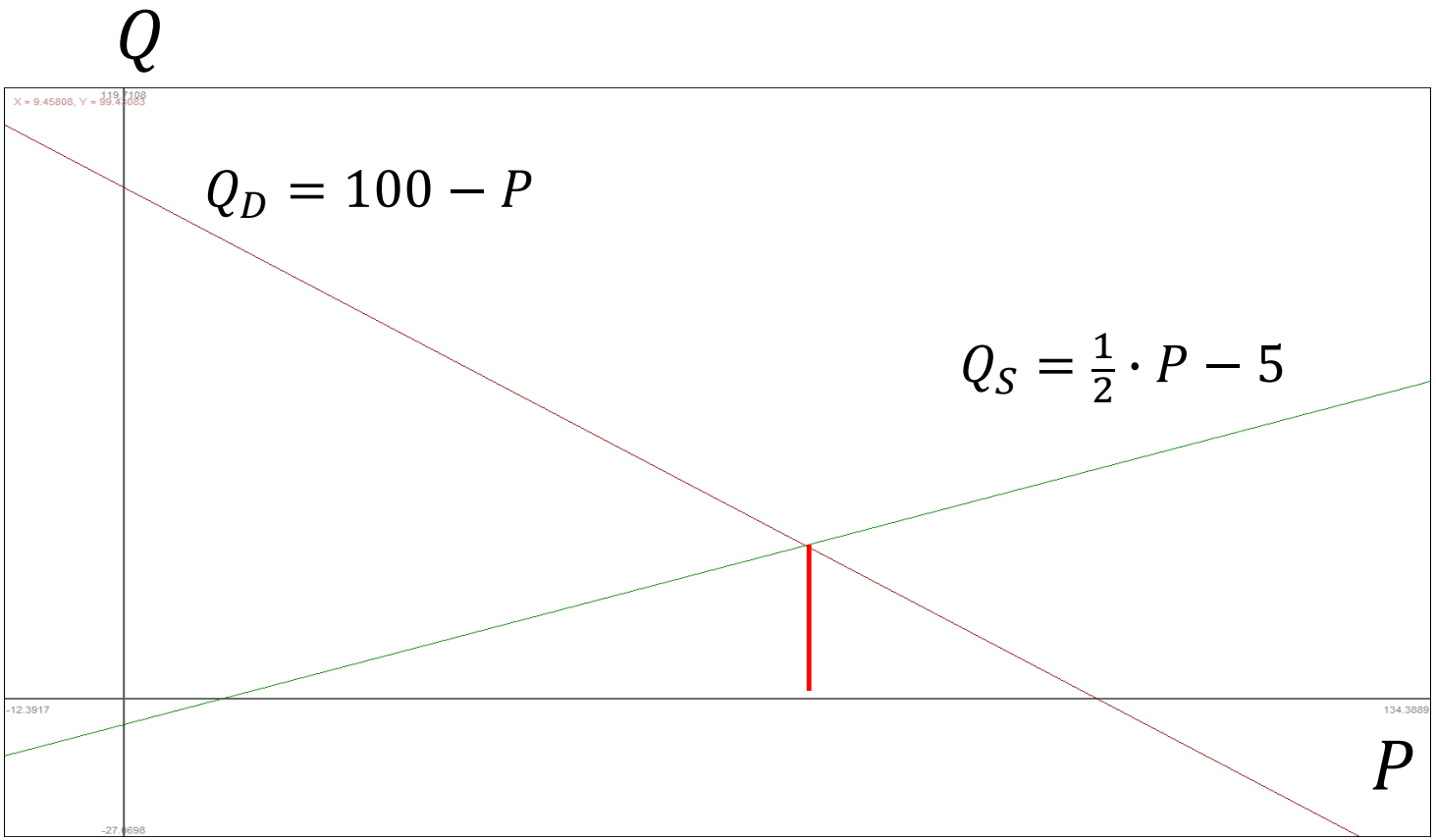
- Suppose demand D for a product depends on its (unit) price P
- Quantity demanded Q_D usually decreases when P increases, e.g.

$$Q_D = 100 - P$$

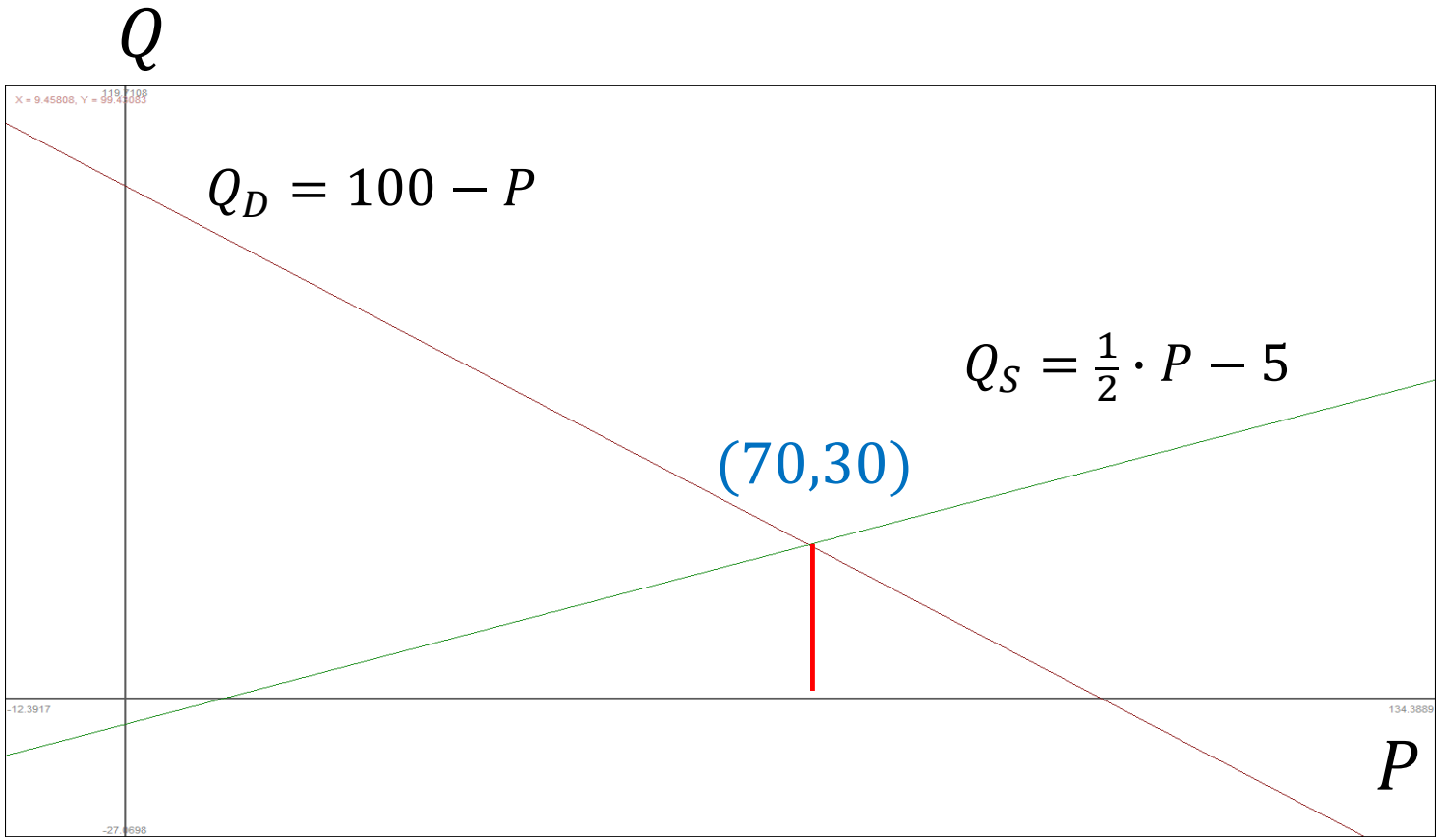
- Suppose supply S also depends on price
- Quantity supplied Q_S usually increases when P increases, e.g.

$$Q_S = \frac{1}{2}P - 5$$

Example - (Linear) Supply & Demand - cont'd



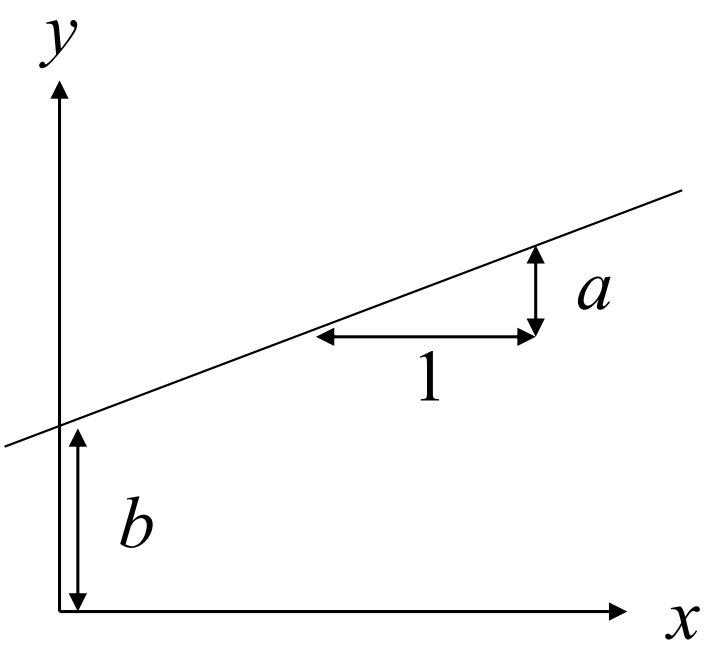
Example - (Linear) Supply & Demand - cont'd



$$Q_D = Q_S \quad \Leftrightarrow \quad 100 - P = \frac{1}{2}P - 5 \quad \Leftrightarrow \quad 105 = \frac{3}{2}P \quad \Leftrightarrow \quad P = 105 \cdot \frac{2}{3} = 70$$

Linear Functions in General

- General shape: $f(x) = ax + b$
- Slope: a $\quad (f'(x) = a)$
- Intercept: b $\quad (f(0) = b)$



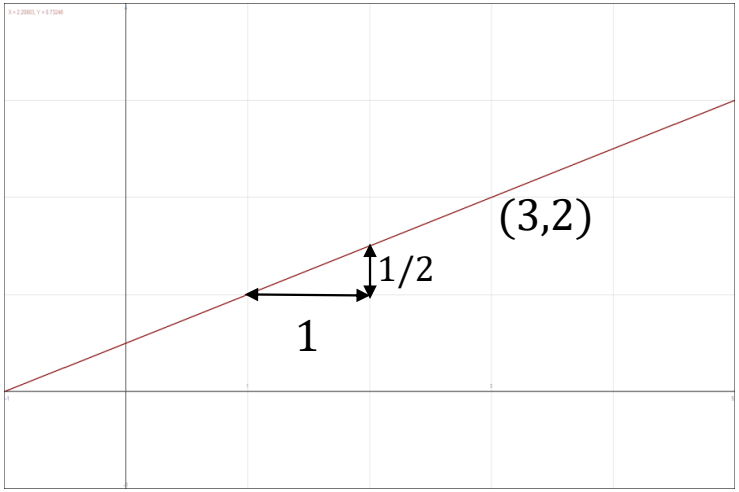
Linear Functions in General - cont'd

Any line ℓ is uniquely pinned down by

(i) ℓ's intercept and ℓ's slope, or

(ii) ℓ's slope and a point on ℓ, or

(iii) two distinct points on ℓ



Quadratic Functions

The general form of a quadratic function is

$$f(x) = ax^2 + bx + c$$

where a, b, c are (real) constants with $a \neq 0$

The graph of a quadratic function is always some kind of parabola. There are three crucial characteristics to it:

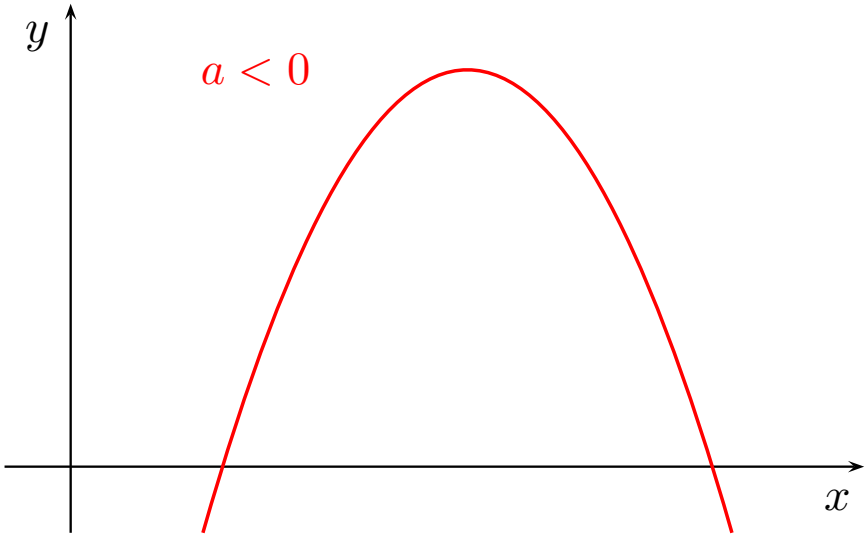
(i) its shape

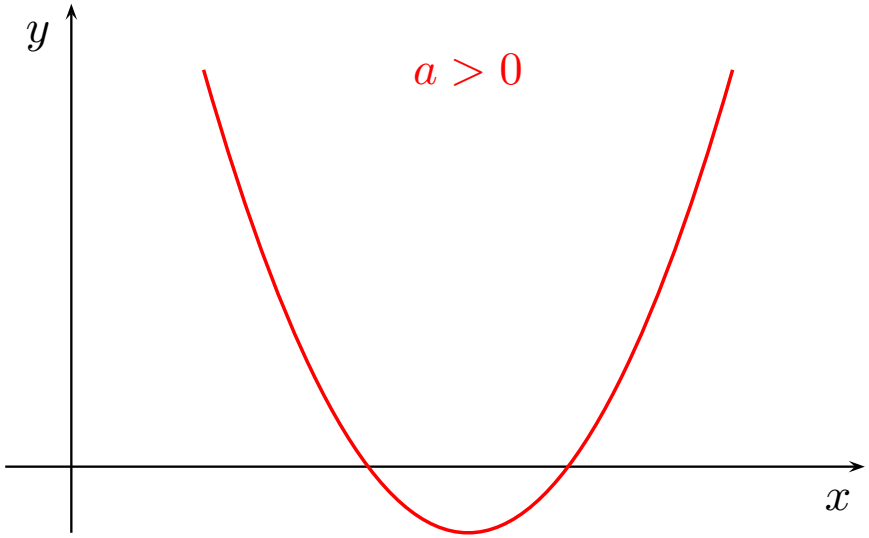
(ii) its extreme point

(iii) its zero(s), i.e., its intersection point(s) with the horizontal axis

Quadratic Functions - Shape

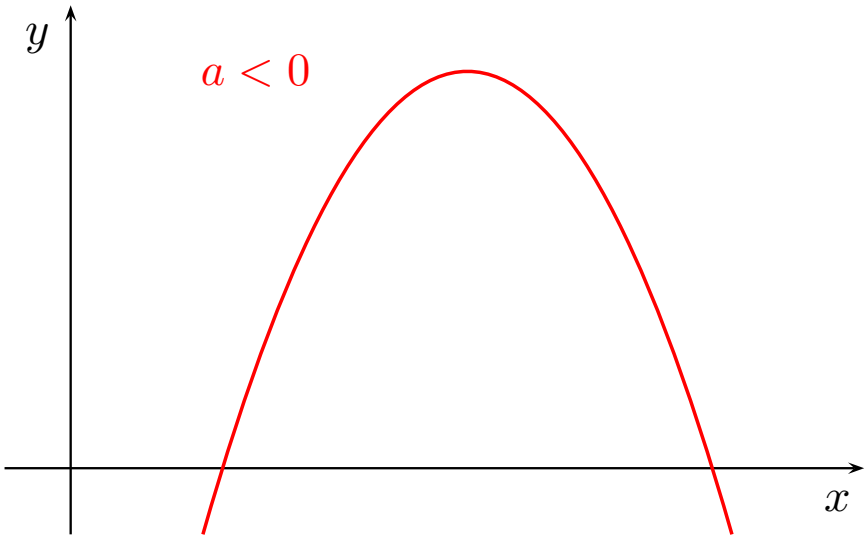
Shape: A parabola $ax^2 + bx + c$ is called concave (it opens downward) if $a < 0$, and it is called convex (it opens upward) if $a > 0$

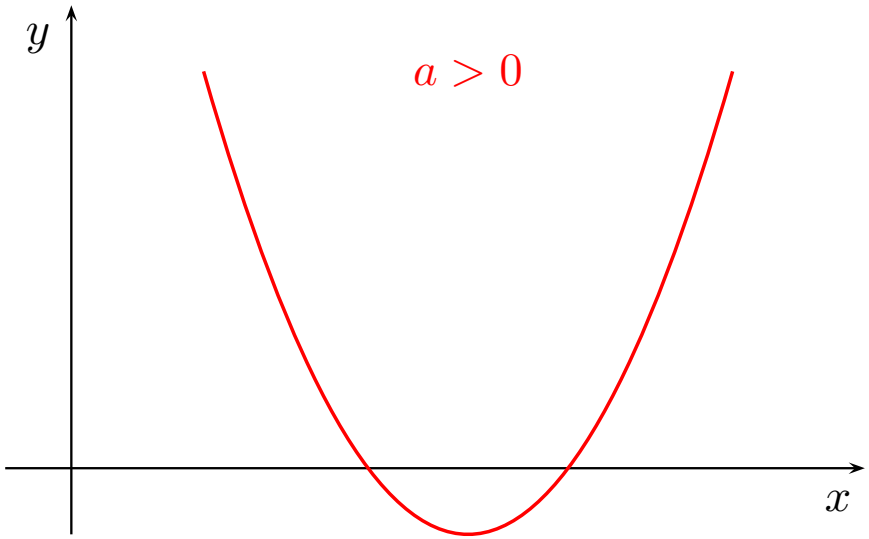




Quadratic Functions - Extreme Points

Extreme Points: The parabola $y = ax^2 + bx + c$ has its etreme point at $x = -\frac{b}{2a}$





Example

Take profits $\pi(P) = -\alpha P^2 + \beta P - c$ only depending on price P. Then, $P = \frac{\beta}{2\alpha}$ maximizes π

Quadratic Functions - Zero(s)

Consider the quadratic equation

$$ax^2 + bx + c = 0 \tag{1}$$

where $a \neq 0$.

Define its so-called discriminant as

$$D = b^2 - 4ac$$

Then:

- if $D > 0$, Equation (1) has two solutions, given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- if $D = 0$, Equation (1) has a single solution, given by $x = \frac{-b}{2a}$
- if $D < 0$, Equation (1) has no solutions

Random Fun Fact

Back to the correlation between math and income

- solving math problems alone can make you rich
- The Clay Mathematics Institute
  - allocates $1 million to the solution of each Millennium Prize Problem
  - six Millennium Prize Problems are still unsolved